Member 2 — Day 3 Documentation

Customer Churn Prediction & Lifetime Value (LTV) Engine

Project Information

Project Name: Customer Churn Prediction & Lifetime Value (LTV) Engine
Team Role: Member 2 — Churn Analysis
Project Day: Day 3
Task Category: Exploratory Data Analysis (EDA)

Day 3 Objective

The primary objective for Member 2 on Day 3 was to perform churn-focused exploratory data analysis (EDA) using the telecom customer dataset.

The analysis focused on identifying customer behavior patterns related to churn and understanding which customer groups are more likely to leave the telecom service.

The work completed today included:

- Understanding churn distribution
- Performing customer behavior analysis
- Creating visualizations using Python libraries
- Saving graphs for future documentation and dashboard usage
- Writing business observations from the analysis
- Organizing notebook structure according to project architecture
- Managing work using Git and GitHub workflow

Folder Navigation Followed

The project structure was maintained according to the master architecture.

Main Project Folder

```
company_name/
└── customer_churn_prediction_ltv_engine/
```

Notebook Location

```
notebooks/eda/
```

Visualization Export Location

```
docs/reports/eda_visualizations/
```

Files Created

Notebook Created

```
notebooks/eda/02_churn_analysis.ipynb
```

Visualization Files Generated

```
docs/reports/eda_visualizations/
├── churn_distribution.png
├── gender_vs_churn.png
├── senior_vs_churn.png
├── internet_service_vs_churn.png
├── payment_method_vs_churn.png
├── monthly_charges_vs_churn.png
└── monthly_charge_histogram.png
```

Technologies and Libraries Used

The following Python libraries and tools were used during the analysis:

Tool/Library	Purpose
Python	Programming language
Pandas	Data loading and manipulation
Matplotlib	Data visualization
Seaborn	Statistical visualization
Jupyter Notebook	Interactive analysis environment
Git	Version control
GitHub	Team collaboration

Dataset Loading and Inspection

The telecom churn dataset was loaded from the raw data folder.

Dataset path:

```
data/raw/telco_customer_churn.csv
```

The dataset was imported using Pandas and inspected using:

- `df.head()`
- `df["Churn"].value_counts()`
- distribution analysis techniques

The churn column was analyzed to determine how many customers stayed and how many customers left the company.

Churn Analysis Performed

Several business-related churn analyses were completed.

1. Churn Distribution Analysis

A count plot was created to understand the overall churn distribution.

Objective

To determine whether most customers stayed or left.

Observation

Most customers remained with the telecom service, but a significant number of customers also churned.

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2. Gender vs Churn Analysis

A churn comparison between male and female customers was performed.

Objective

To identify whether gender influences churn behavior.

Observation

Gender did not appear to have a strong impact on churn behavior.

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3. Senior Citizen vs Churn Analysis

Senior citizen customers were analyzed against churn patterns.

Objective

To identify whether older customers are more likely to churn.

Observation

Senior citizen customers showed relatively higher churn behavior compared to non-senior customers.

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4. Internet Service vs Churn Analysis

Different internet service types were compared with churn.

Services Analyzed

- DSL
- Fiber Optic
- No Internet Service

Objective

To identify which internet service category experiences the highest churn.

Observation

Fiber Optic customers appeared to churn more frequently than DSL customers.

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5. Payment Method vs Churn Analysis

Customer payment methods were analyzed against churn.

Objective

To identify risky payment behaviors.

Observation

Customers using electronic check payment methods appeared to have higher churn rates.

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6. Monthly Charges vs Churn Analysis

Monthly charges were compared against churn behavior using boxplots and histograms.

Objective

To determine whether expensive plans influence churn.

Observation

Customers with higher monthly charges were more likely to churn.

Visualization Export Process

All generated charts were exported and saved inside:

```
docs/reports/eda_visualizations/
```

The `plt.savefig()` function was used to save images.

An automatic directory creation approach was also implemented using:

```
import os
os.makedirs("../../docs/reports/eda_visualizations", exist_ok=True)
```

This ensured the folder was automatically created if missing.

Notebook Documentation Practices

Markdown cells were added throughout the notebook to explain:

- graph purpose
- business meaning
- churn behavior patterns
- observations from visual analysis

This improved notebook readability and professional documentation quality.

Git and GitHub Workflow

A professional feature branch workflow was followed.

Branch Created

```
git checkout -b feature/churn-analysis-member2
```

Git Operations Performed

- repository initialization
- git status checking
- staging files using git add
- committing work using git commit
- branch-based workflow management

Challenges Faced and Solutions

Several beginner-level issues were encountered and resolved during implementation.

Issue	Solution
Jupyter not recognized	Installed notebook package using pip
Graphs not saving	Created visualization folder correctly
Git tracking system folders	Reinitialized Git properly inside project
Git author identity error	Configured Git username and email
Missing visualization directory	Created directory manually and using os.makedirs()

Key Business Insights Identified

The analysis produced several important churn-related insights:

- Customers with Fiber Optic internet services tend to churn more.
- Customers paying through electronic checks show higher churn behavior.
- Higher monthly charges are associated with increased churn.
- Senior citizens appear to have relatively higher churn rates.
- Gender does not significantly affect churn.

These findings will help future stages of the project including:

- feature engineering
- churn prediction modeling
- customer segmentation
- dashboard reporting
- business recommendations

Final Outcome

By the end of Day 3, the following deliverables were successfully completed:

- Churn analysis notebook created
- Multiple business-focused visualizations generated
- Visualizations exported successfully
- Observations documented professionally
- Git workflow implemented using feature branches
- Project folder structure maintained correctly

The work completed today contributes directly to future machine learning model development and business intelligence reporting within the Customer Churn Prediction & Lifetime Value Engine project.

Final Directory Structure

```
customer_churn_prediction_ltv_engine/
│
├── notebooks/
│   └── eda/
│       └── 02_churn_analysis.ipynb
│
├── docs/
│   └── reports/
│       └── eda_visualizations/
│           ├── churn_distribution.png
│           ├── gender_vs_churn.png
│           ├── senior_vs_churn.png
│           ├── internet_service_vs_churn.png
│           ├── payment_method_vs_churn.png
│           ├── monthly_charges_vs_churn.png
│           └── monthly_charge_histogram.png
```